

Supplementary Material

Step-by-step derivations for “Diffusion anisotropy and entropy: reflections of each other”

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This document derives every relation stated in the paper, one step at a time, in the paper’s notation. Equation numbers without an S refer to the paper. Each derivation ends with the name of the check in `analysis/identity_checks.py` of the code repository that verifies it numerically.

S1 Setup

Let $x_1, \dots, x_K$ be positive numbers. Their mean is $\bar{x} = \frac{1}{K} \sum_i x_i$. The average of any quantity over i is written $\langle \cdot \rangle$, so $\bar{x} = \langle x \rangle$. The standard deviation uses divisor K ,

$$\sigma_x^2 = \frac{1}{K} \sum_i (x_i - \bar{x})^2 = \langle x^2 \rangle - \bar{x}^2, \quad (\text{S1})$$

and the coefficient of variation is $\text{CV}_x = \sigma_x / \bar{x}$. Dividing Eq. (S1) by $\bar{x}^2$ gives

$$\frac{\langle x^2 \rangle}{\bar{x}^2} = 1 + \text{CV}_x^2. \quad (\text{S2})$$

The proportions are $p_i = x_i / \sum_j x_j$. They are positive and sum to 1, so they form a probability distribution. When all values are equal, every p_i equals $1/K$. This is the uniform distribution u , with $u_i = 1/K$. Because $\sum_j x_j = K\bar{x}$,

$$p_i = \frac{x_i}{K\bar{x}}. \quad (\text{S3})$$

The Rényi entropy of order $\alpha \neq 1$ is $H_\alpha(p) = (1 - \alpha)^{-1} \ln \sum_i p_i^\alpha$. The Shannon entropy $H_1 = -\sum_i p_i \ln p_i$ is its limit as $\alpha \rightarrow 1$ (Section S4). The power mean of order $r \neq 0$ is $M_r(x) = \langle x^r \rangle^{1/r}$, and $M_0 = \exp(\ln x)$ is the geometric mean. $M_1 = \bar{x}$. The power mean increases with r , and all power means are equal only when all values are equal (the power-mean inequality).

S2 The Jensen gap

The power-mean Jensen gap of order α (Eq. 1) is

$$J_\alpha(x) = \frac{\langle x^\alpha \rangle}{\bar{x}^\alpha} - 1. \quad (\text{S4})$$

Jensen’s inequality says that for a convex function φ , $\langle \varphi(x) \rangle \geq \varphi(\bar{x})$, with equality only when all values are equal. For a concave function the inequality reverses. The function $\varphi(x) = x^\alpha$ is convex for $\alpha > 1$ and for $\alpha < 0$, and concave for $0 < \alpha < 1$. So $\langle x^\alpha \rangle - \bar{x}^\alpha$, and with it J_α , is positive for $\alpha > 1$ and $\alpha < 0$ and negative for $0 < \alpha < 1$, unless all values are equal. In both cases J_α has the sign of $\alpha(\alpha - 1)$.

Since $\langle x^\alpha \rangle = M_\alpha^\alpha$ and $\bar{x} = M_1$, the gap is also

$$1 + J_\alpha = \left(\frac{M_\alpha}{M_1} \right)^\alpha. \quad (\text{S5})$$

At order 2, Eq. (S2) gives

$$J_2 = \frac{\langle x^2 \rangle}{\bar{x}^2} - 1 = \text{CV}_x^2. \quad (\text{S6})$$

S3 The identity, Eq. 2

Step 1, the power sum. Raise Eq. (S3) to the power α and sum over i :

$$\sum_i p_i^\alpha = \frac{\sum_i x_i^\alpha}{(K\bar{x})^\alpha} = \frac{K\langle x^\alpha \rangle}{K^\alpha \bar{x}^\alpha} = K^{1-\alpha} \frac{\langle x^\alpha \rangle}{\bar{x}^\alpha} = K^{1-\alpha} (1 + J_\alpha). \quad (\text{S7})$$

Step 2, the entropy. Take $(1 - \alpha)^{-1} \ln$ of Eq. (S7):

$$H_\alpha = \frac{1}{1 - \alpha} \left[(1 - \alpha) \ln K + \ln(1 + J_\alpha) \right] = \ln K + \frac{\ln(1 + J_\alpha)}{1 - \alpha}. \quad (\text{S8})$$

Step 3, the entropy deficit. Move $\ln K$ to the left:

$$\ln K - H_\alpha = \frac{\ln(1 + J_\alpha)}{\alpha - 1}. \quad (\text{S9})$$

Step 4, the divergence. The Rényi divergence of p from u is $\text{Div}_\alpha(p \| u) = (\alpha - 1)^{-1} \ln \sum_i p_i^\alpha u_i^{1-\alpha}$. Since $u_i^{1-\alpha} = K^{\alpha-1}$ for every i ,

$$\text{Div}_\alpha(p \| u) = \frac{1}{\alpha - 1} \left[(\alpha - 1) \ln K + \ln \sum_i p_i^\alpha \right] = \ln K - \frac{\ln \sum_i p_i^\alpha}{1 - \alpha} = \ln K - H_\alpha. \quad (\text{S10})$$

Step 5, the power-mean form. Insert Eq. (S5) into Eq. (S9):

$$\ln K - H_\alpha = \frac{1}{\alpha - 1} \ln \left(\frac{M_\alpha}{M_1} \right)^\alpha = \frac{\alpha}{\alpha - 1} \ln \frac{M_\alpha}{M_1}. \quad (\text{S11})$$

Steps 3 to 5 together are Eq. 2.

Step 6, the sign. For $\alpha > 1$, $M_\alpha \geq M_1$ and $\alpha/(\alpha - 1) > 0$, so the deficit is nonnegative. For $0 < \alpha < 1$, $M_\alpha \leq M_1$, so the logarithm is nonpositive, and $\alpha/(\alpha - 1) < 0$, so the deficit is again nonnegative. In both cases it is zero only when all values are equal.

Step 7, monotonicity. For fixed α , the right side of Eq. (S9) is a strictly monotone function of J_α , increasing for $\alpha > 1$ and decreasing for $\alpha < 1$. Knowing one of the entropy deficit and the Jensen gap of the same order therefore fixes the other.

Numerical check: Eq. 2 identity, every form, all orders | Eq. 2 derivation: $\sum p^\alpha = K^{1-\alpha}(1 + J_\alpha)$

S4 Order 1, the Shannon case

The limit. At $\alpha = 1$ both $\ln \sum_i p_i^\alpha$ and $1 - \alpha$ vanish. By l'Hôpital's rule, with $\frac{d}{d\alpha} \sum_i p_i^\alpha = \sum_i p_i^\alpha \ln p_i$ and $\sum_i p_i = 1$,

$$H_1 = \lim_{\alpha \rightarrow 1} \frac{\ln \sum_i p_i^\alpha}{1 - \alpha} = \frac{\sum_i p_i \ln p_i}{-1} = - \sum_i p_i \ln p_i. \quad (\text{S12})$$

The deficit is the Kullback–Leibler divergence. Since $\sum_i p_i = 1$,

$$\ln K - H_1 = \sum_i p_i \ln K + \sum_i p_i \ln p_i = \sum_i p_i \ln \frac{p_i}{1/K} = \text{Div}_{\text{KL}}(p \parallel u). \quad (\text{S13})$$

In terms of the values. By Eq. (S3), $Kp_i = x_i/\bar{x}$, so

$$\ln K - H_1 = \sum_i p_i \ln \frac{x_i}{\bar{x}} = \sum_i p_i \ln x_i - \ln \bar{x}. \quad (\text{S14})$$

This is Theil's index of the shares p_i .

The Jensen gap of $x \ln x$. For $\varphi(x) = x \ln x$, again using Eq. (S3),

$$\frac{\langle x \ln x \rangle - \bar{x} \ln \bar{x}}{\bar{x}} = \sum_i \frac{x_i}{K\bar{x}} \ln x_i - \ln \bar{x} = \sum_i p_i \ln x_i - \ln \bar{x}, \quad (\text{S15})$$

which is Eq. (S14).

The reverse direction. With the roles of p and u exchanged,

$$\text{Div}_{\text{KL}}(u \parallel p) = \sum_i \frac{1}{K} \ln \frac{1/K}{p_i} = - \left\langle \ln \frac{x}{\bar{x}} \right\rangle = \ln \bar{x} - \langle \ln x \rangle = \ln \frac{M_1}{M_0}. \quad (\text{S16})$$

It can also be written $\text{Div}_{\text{KL}}(u \parallel p) = H(u, p) - \ln K$, where $H(u, p) = - \sum_i u_i \ln p_i$ is the cross-entropy of u relative to p , not an entropy of p .

Numerical check: `uniform log gap ln(mean) - mean(ln) = KL(u||p) | Shannon deficit = Jensen gap of x ln x, scaled by mean`

S5 Other special orders

Order ∞ . As $\alpha \rightarrow \infty$, the largest proportion dominates $\sum_i p_i^\alpha$. If m proportions share the largest value $p_{\max}$, then $\sum_i p_i^\alpha = m p_{\max}^\alpha (1 + o(1))$, and

$$H_\infty = \lim_{\alpha \rightarrow \infty} \frac{\ln m + \alpha \ln p_{\max}}{1 - \alpha} = - \ln p_{\max}, \quad \ln K - H_\infty = \ln(K p_{\max}) = \ln \frac{x_{\max}}{\bar{x}}. \quad (\text{S17})$$

Numerical check: `order inf: ln K - H_inf = ln(max x / mean x)`

Order 2. Eq. (S7) with $\alpha = 2$ and Eq. (S6) give $\sum_i p_i^2 = (1 + \text{CV}_x^2)/K$. Then

$$\ln K - H_2 = \ln(1 + \text{CV}_x^2), \quad \text{PR} = \frac{1}{\sum_i p_i^2} = \frac{K}{1 + \text{CV}_x^2}, \quad \frac{\text{PR}}{K} = \frac{1}{1 + \text{CV}_x^2}. \quad (\text{S18})$$

The chi-squared divergence. Expanding the square and using $\sum_i p_i = 1$,

$$\sum_i \frac{(p_i - u_i)^2}{u_i} = K \sum_i \left(p_i^2 - \frac{2p_i}{K} + \frac{1}{K^2} \right) = K \sum_i p_i^2 - 2 + 1 = (1 + \text{CV}_x^2) - 1 = \text{CV}_x^2. \quad (\text{S19})$$

Numerical check: order 2: PR/K = 1/(1+CV^2) | order 2: CV^2 = chi^2(p||u)

S6 The geometric form, Eq. 3

Let θ be the angle between the vector $x = (x_1, \dots, x_K)$ and the isotropic direction $\mathbf{1} = (1, \dots, 1)$. Then

$$\cos^2 \theta = \frac{(x \cdot \mathbf{1})^2}{|x|^2 |\mathbf{1}|^2} = \frac{(\sum_i x_i)^2}{K \sum_i x_i^2} = \frac{K^2 \bar{x}^2}{K \cdot K \langle x^2 \rangle} = \frac{\bar{x}^2}{\langle x^2 \rangle} = \frac{1}{1 + \text{CV}_x^2} = \frac{\text{PR}}{K}, \quad (\text{S20})$$

using Eqs. (S2) and (S18). Then $\tan^2 \theta = 1/\cos^2 \theta - 1 = \text{CV}_x^2$, and since the values are positive, $0 \leq \theta < \pi/2$ and $\tan \theta = \text{CV}_x$. Also $\sin^2 \theta = 1 - \cos^2 \theta = \text{CV}_x^2/(1 + \text{CV}_x^2)$.

Numerical check: PR/K = cos^2 theta | CV = tan theta

S7 Near isotropy, Eq. 4

Write $x_i = \bar{x}(1 + \varepsilon_i)$. Averaging gives $\langle \varepsilon \rangle = 0$, and $\langle \varepsilon^2 \rangle = \sigma_x^2/\bar{x}^2 = \text{CV}_x^2$. Expanding $(1 + \varepsilon_i)^\alpha$ to second order,

$$1 + J_\alpha = \langle (1 + \varepsilon)^\alpha \rangle = 1 + \alpha \langle \varepsilon \rangle + \frac{1}{2} \alpha(\alpha - 1) \langle \varepsilon^2 \rangle + O(\varepsilon^3) = 1 + \frac{1}{2} \alpha(\alpha - 1) \text{CV}_x^2 + O(\varepsilon^3). \quad (\text{S21})$$

Since $\ln(1 + J) = J + O(J^2)$, Eq. (S9) gives

$$\ln K - H_\alpha = \frac{J_\alpha}{\alpha - 1} + O(\varepsilon^3) = \frac{\alpha}{2} \text{CV}_x^2 + O(\varepsilon^3). \quad (\text{S22})$$

The coefficient of the cubic term grows with α , so the approximation is not uniform in the order. At $\alpha = \infty$ the deficit $\ln(x_{\max}/\bar{x}) = \ln(1 + \varepsilon_{\max}) \approx \varepsilon_{\max}$ is first order.

Numerical check: second-order form overstates the order-2 member at CV 0.544 (fraction)

S8 Divergences are not metrics

A metric must be symmetric and satisfy the triangle inequality. At order $\frac{1}{2}$, $\text{Div}_{1/2}(p \| q) = -2 \ln \sum_i \sqrt{p_i q_i}$, which is symmetric in p and q . It fails the triangle inequality. For two outcomes, take $p = (0.9, 0.1)$, $q = (0.5, 0.5)$ and $r = (0.1, 0.9)$. Then $\text{Div}_{1/2}(p \| q) = \text{Div}_{1/2}(q \| r) = 0.2231$, but $\text{Div}_{1/2}(p \| r) = -2 \ln 0.6 = 1.0217$, which exceeds their sum. Other positive orders are not symmetric. For example, at order 1, $\text{Div}_{\text{KL}}(p \| u) \neq \text{Div}_{\text{KL}}(u \| p)$ in general, as Section S4 shows.

Numerical check: order 1/2 is symmetric | orders 1, 2 and 4 are asymmetric | order 1/2 fails the triangle inequality | Supplement: triangle example

S9 The diffusion tensor

Take $x = \lambda$, the eigenvalues of the tensor, with $K = 3$, mean $\bar{\lambda}$ and normalized eigenvalues $p_i^{(\lambda)} = \lambda_i / \sum_j \lambda_j$.

FA, Eq. 5. Fractional anisotropy is $\text{FA}^2 = \frac{3}{2} \sum_i (\lambda_i - \bar{\lambda})^2 / \sum_i \lambda_i^2$. With divisor 3, $\sum_i (\lambda_i - \bar{\lambda})^2 = 3\sigma_\lambda^2 = 3\bar{\lambda}^2 \text{CV}_\lambda^2$ and, by Eq. (S2), $\sum_i \lambda_i^2 = 3\langle \lambda^2 \rangle = 3\bar{\lambda}^2(1 + \text{CV}_\lambda^2)$. So

$$\text{FA}^2 = \frac{3}{2} \frac{\text{CV}_\lambda^2}{1 + \text{CV}_\lambda^2}. \quad (\text{S23})$$

By Eq. (S18), $\text{Div}_2 = \ln 3 - H_2 = \ln(1 + \text{CV}_\lambda^2)$, so $e^{-\text{Div}_2} = 1/(1 + \text{CV}_\lambda^2)$ and $1 - e^{-\text{Div}_2} = \text{CV}_\lambda^2/(1 + \text{CV}_\lambda^2)$. Hence

$$\text{FA}^2 = \frac{3}{2} (1 - e^{-\text{Div}_2}) = \frac{3}{2} (1 - \text{PR}_\lambda/3) = \frac{3}{2} \sin^2 \theta, \quad (\text{S24})$$

since $\text{PR}_\lambda/3 = e^{-\text{Div}_2}$ (Eq. (S18)) and $\sin^2 \theta = \text{CV}_\lambda^2/(1 + \text{CV}_\lambda^2)$.

FA and the order-2 entropy. From $\text{FA}^2 = \frac{3}{2}(1 - e^{H_2}/3)$, solve for H_2 : $e^{H_2} = 3 - 2\text{FA}^2$, so

$$H_2 = \ln(3 - 2\text{FA}^2), \quad \ln 3 - H_2 = -\ln(1 - \frac{2}{3}\text{FA}^2). \quad (\text{S25})$$

Near isotropy the second form is $\frac{2}{3}\text{FA}^2$ to leading order.

Numerical check: tensor orders, FA, QA and VR relations | $\ln 3 - H_2(\lambda) = -\ln(1 - 2\text{FA}^2/3)$

The trace form of Özarslan et al. The trace-normalized tensor $R = \mathbf{D}/\text{tr}\mathbf{D}$ has eigenvalues $p_i^{(\lambda)}$, so $\text{tr}R^2 = \sum_i (p_i^{(\lambda)})^2 = e^{-H_2}$. Then $\text{FA}^2 = \frac{3}{2}(1 - 1/(3\text{tr}R^2))$, which is $\text{FA} = \sqrt{(3 - 1/\text{tr}R^2)/2}$.

Numerical check: Özarslan Eq. 11, FA from trace(R^2), and trace(R^2) = exp(-H_2)

Procrustes anisotropy. Dryden et al. define PA as FA computed on $q_i = \sqrt{\lambda_i}$. Since $\sum_i (q_i - \bar{q})^2 = \sum_i q_i^2 - 3\bar{q}^2$ and $\sum_i q_i^2 = \sum_i \lambda_i$,

$$\text{PA}^2 = \frac{3}{2} \left(1 - \frac{(\sum_i q_i)^2}{3 \sum_i q_i^2} \right). \quad (\text{S26})$$

At order $\frac{1}{2}$, with $u_i^{1/2} = 3^{-1/2}$, $\text{Div}_{1/2} = -2 \ln \sum_i \sqrt{p_i^{(\lambda)}}/3 = \ln 3 - 2 \ln \sum_i \sqrt{p_i^{(\lambda)}}$, so $e^{-\text{Div}_{1/2}} = (\sum_i \sqrt{p_i^{(\lambda)}})^2/3$. Since $\sqrt{p_i^{(\lambda)}} = q_i/\sqrt{\sum_j \lambda_j}$, $(\sum_i \sqrt{p_i^{(\lambda)}})^2 = (\sum_i q_i)^2/\sum_i q_i^2$. Therefore $\text{PA}^2 = \frac{3}{2}(1 - e^{-\text{Div}_{1/2}})$.

Numerical check: Procrustes anisotropy (Dryden 2009): $\text{PA}^2 = (3/2)(1 - \exp(-\text{Div}_{1/2}))$

Relative anisotropy. In one common convention $\text{RA} = \sqrt{\sum_i (\lambda_i - \bar{\lambda})^2 / (\sqrt{3} \bar{\lambda})} = \sigma_\lambda / \bar{\lambda} = \text{CV}_\lambda$. Other conventions multiply this by a constant.

Quantitative anisotropy, Eq. 6. By Eq. (S14) with $x = \lambda$, $\ln 3 - H_1(p^{(\lambda)}) = \sum_i p_i^{(\lambda)} \ln \lambda_i - \ln \bar{\lambda}$. The von Neumann entropy of R is $-\text{tr}(R \ln R) = -\sum_i p_i^{(\lambda)} \ln p_i^{(\lambda)} = H_1(p^{(\lambda)})$, because R has eigenvalues $p_i^{(\lambda)}$. So the order-1 member is $\ln 3$ minus the von Neumann entropy of R .

The volume ratio. $\text{VR} = \lambda_1 \lambda_2 \lambda_3 / \bar{\lambda}^3$. The geometric mean is $M_0 = (\lambda_1 \lambda_2 \lambda_3)^{1/3}$, so $\text{VR} = (M_0/M_1)^3$, and by Eq. (S16)

$$-\frac{1}{3} \ln \text{VR} = \ln \frac{M_1}{M_0} = \text{Div}_{\text{KL}}(u \| p^{(\lambda)}). \quad (\text{S27})$$

FA and tensor mode fix the normalized eigenvalues. The normalized eigenvalues sum to 1, so they have two degrees of freedom. Write $d = p^{(\lambda)} - u$. Its components sum to zero, so d lies in a plane, and d is orthogonal to u . Hence $\sum_i (p_i^{(\lambda)})^2 = |u|^2 + |d|^2 = \frac{1}{3} + |d|^2$, and $|d|$ fixes $\sum_i (p_i^{(\lambda)})^2$ and therefore FA. In the plane, use the orthonormal basis $e_1 = (2, -1, -1)/\sqrt{6}$, $e_2 = (0, 1, -1)/\sqrt{2}$ and write $d = |d|(\cos \phi e_1 + \sin \phi e_2)$. The product of the components of d is $|d|^3 \cos 3\phi / (3\sqrt{6})$, so the tensor mode, $3\sqrt{6} \prod_i d_i / |d|^3$, equals $\cos 3\phi$. With $\lambda_1 \geq \lambda_2 \geq \lambda_3$, ϕ runs from 0 (linear, mode +1) to $\pi/3$ (planar, mode -1), and $\cos 3\phi$ is one-to-one on that range. So FA fixes $|d|$, mode fixes ϕ , and together they fix $p^{(\lambda)}$ and every entropy of it.

Numerical check: Supplement: mode = cos 3 phi and sum p^2 = 1/3 + |d|^2

S10 The directional profile

GFA, Eq. 7. For N values D_i , Tuch's generalized fractional anisotropy is the standard deviation, with divisor $N - 1$, over the root mean square:

$$\text{GFA}^2 = \frac{N \sum_i (D_i - \bar{D})^2}{(N - 1) \sum_i D_i^2} = \frac{N}{N - 1} \frac{N \sigma_D^2}{N \langle D^2 \rangle} = \frac{N}{N - 1} \frac{\text{CV}_D^2}{1 + \text{CV}_D^2}. \quad (\text{S28})$$

By Eq. (S18) and the geometric form this equals $\frac{N}{N-1}(1 - \text{PR}/N) = \frac{N}{N-1} \sin^2 \theta$. With $N = 3$ the prefactor is $\frac{3}{2}$ and this is Eq. 5.

The CVD of Aja-Fernández et al. With $\log E_i = -bD_i$, their sample variance and second moment are

$$\mathcal{V} = \frac{N}{N-1} [\langle (\log E)^2 \rangle - \langle \log E \rangle^2] = \frac{N}{N-1} b^2 \sigma_D^2, \quad \langle (\log E)^2 \rangle = b^2 \langle D^2 \rangle. \quad (\text{S29})$$

So $\text{CVD}^2 = \mathcal{V} / \langle (\log E)^2 \rangle = \frac{N}{N-1} \sigma_D^2 / \langle D^2 \rangle = \text{GFA}^2$.

Numerical check: CVD^2 of Aja-Fernandez 2018 = GFA^2 of the profile, Eq. (7)

The signal domain. The signals $S_i/S_0 = e^{-bD_i}$ are positive, so the identity applies to them as well. Near isotropy write $D_i = \bar{D} + \delta_i$ with $\langle \delta \rangle = 0$ and $\langle \delta^2 \rangle = \sigma_D^2$. Then $e^{-bD_i} = e^{-b\bar{D}} e^{-b\delta_i} \approx e^{-b\bar{D}} (1 - b\delta_i)$. The signals have mean $\approx e^{-b\bar{D}}$ and standard deviation $\approx e^{-b\bar{D}} b \sigma_D$, so their coefficient of variation is $\approx b \sigma_D$. By Section S7 the entropy deficits of the two domains are $\approx \frac{\alpha}{2} \text{CV}_D^2$ and $\approx \frac{\alpha}{2} (b \sigma_D)^2$, and their ratio is

$$\frac{(b \sigma_D)^2}{(\sigma_D / \bar{D})^2} = (b \bar{D})^2. \quad (\text{S30})$$

Numerical check: Sec 4.1 near isotropy: signal/diffusivity deficit ratio vs (b Dbar)^2

S11 The link to the tensor, Eq. 8

For a single Gaussian compartment the profile is $D(\hat{g}) = \hat{g}^\top \mathbf{D} \hat{g}$. Averages over the unit sphere of products of the components of $\hat{g}$ are

$$\langle g_i g_j \rangle = \frac{1}{3} \delta_{ij}, \quad \langle g_i g_j g_k g_l \rangle = \frac{1}{15} (\delta_{ij} \delta_{kl} + \delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}). \quad (\text{S31})$$

The mean of the profile is $\langle D \rangle = \sum_{ij} D_{ij} \langle g_i g_j \rangle = \text{tr} \mathbf{D} / 3 = \bar{\lambda}$. Its mean square is

$$\langle D^2 \rangle = \sum_{ijkl} D_{ij} D_{kl} \langle g_i g_j g_k g_l \rangle = \frac{1}{15} [(\text{tr} \mathbf{D})^2 + 2 \text{tr} \mathbf{D}^2] = \frac{1}{15} [9 \bar{\lambda}^2 + 2 \sum_i \lambda_i^2]. \quad (\text{S32})$$

Its variance is

$$\langle D^2 \rangle - \langle D \rangle^2 = \frac{1}{15} [2 \sum_i \lambda_i^2 - 6 \bar{\lambda}^2] = \frac{2}{15} \sum_i (\lambda_i - \bar{\lambda})^2 = \frac{2}{5} \sigma_\lambda^2, \quad (\text{S33})$$

using $\sum_i (\lambda_i - \bar{\lambda})^2 = \sum_i \lambda_i^2 - 3 \bar{\lambda}^2 = 3 \sigma_\lambda^2$. Dividing by $\bar{\lambda}^2$,

$$\text{CV}_D^2 = \frac{2}{5} \text{CV}_\lambda^2. \quad (\text{S34})$$

Equal-weight averages over N near-uniform directions approximate these sphere averages.

The linear limit. For $\lambda \propto (1, 0, 0)$, $\text{CV}_\lambda^2 = 2$ and $\text{CV}_D^2 = \frac{4}{5}$. The normalized second moments are $1 + \text{CV}_\lambda^2 = 3$ and $1 + \text{CV}_D^2 = \frac{9}{5}$, whose ratio is $\frac{3}{5}$, the value Özarslan et al. report for the profile against the eigenvalues.

Numerical check: `CV_D^2 / CV_lambda^2` on a dense sphere, 50 random tensors | linear tensor: profile's normalized second moment / eigenvalues' = 3/5

S12 The order-1 link between profile and tensor

In the eigenvector basis, $D(\hat{g}) = \sum_i w_i(\hat{g}) \lambda_i$ with $w_i(\hat{g}) = (\hat{g} \cdot e_i)^2 \geq 0$ and $\sum_i w_i = 1$. So each direction's diffusivity is a weighted mean of the eigenvalues. Let $f(x) = x \ln x$, which is convex. Jensen's inequality for each direction gives $f(D(\hat{g})) \leq \sum_i w_i(\hat{g}) f(\lambda_i)$. Averaging over the sphere, with $\langle w_i \rangle = \frac{1}{3}$,

$$\langle f(D) \rangle \leq \frac{1}{3} \sum_i f(\lambda_i) = \langle f(\lambda) \rangle. \quad (\text{S35})$$

The profile and the eigenvalues have the same mean $\bar{\lambda}$, so subtracting $f(\bar{\lambda})$ and dividing by $\bar{\lambda}$ turns both sides into order-1 deficits (Section S4). The profile's order-1 deficit is therefore at most the eigenvalues', with equality only at isotropy. Near isotropy both deficits are $\approx \frac{1}{2} \text{CV}^2$, so their ratio tends to $\text{CV}_D^2 / \text{CV}_\lambda^2 = \frac{2}{5}$.

Numerical check: order-1 link: Jensen violations over the eigenvalue simplex | order-1 link: median ratio near isotropy (FA < 0.05)

S13 The chain rule at order 1

Split the K values into groups g . Let $P_g = \sum_{i \in g} p_i$ be a group's proportion of the total and $U_g = |g|/K$ its fraction of the values. Within a group, $p_{i|g} = p_i/P_g$ and $u_{i|g} = 1/|g|$, so $p_i = P_g p_{i|g}$ and $u_i = U_g u_{i|g}$. Then

$$\text{Div}_{\text{KL}}(p \parallel u) = \sum_g \sum_{i \in g} P_g p_{i|g} \left[\ln \frac{P_g}{U_g} + \ln \frac{p_{i|g}}{u_{i|g}} \right] = \text{Div}_{\text{KL}}(P \parallel U) + \sum_g P_g \text{Div}_{\text{KL}}(p_{\cdot|g} \parallel u_{\cdot|g}), \quad (\text{S36})$$

using $\sum_{i \in g} p_{i|g} = 1$. The order-1 deficit is the proportion-weighted mean of the within-group deficits plus the divergence of the groups' proportions from their fractions of the values.

Numerical check: order-1 deficit = share-weighted within-group deficits + between-group divergence